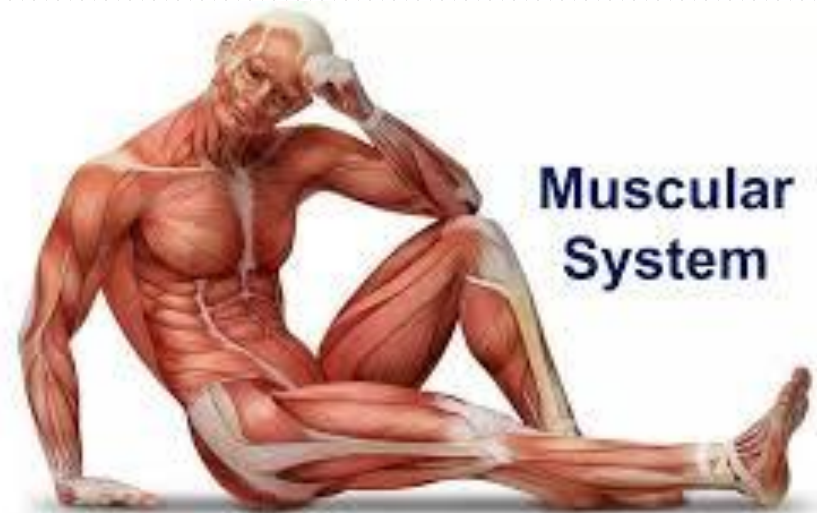


Management of patient with Musculoskeletal diseases



**Muscular
System**

Dr . Bardies Shaban



الحديث الجانبي يشتت الانتباه



من فضلك اغلق الهاتف

Fracture

Hip
replacement

Rheumatic
arthritis

Osteomyelitis

Osteoporosis



1- Fracture

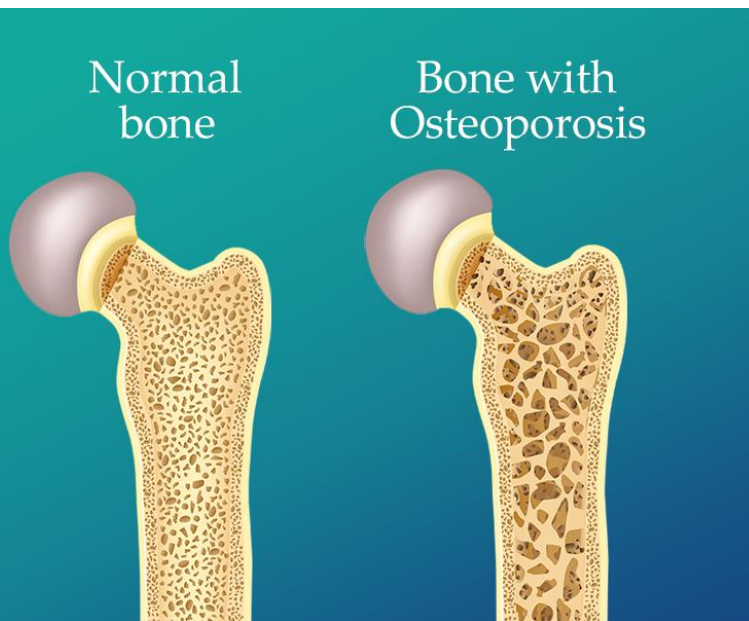
- ◉ Definition
- ◉ Causes
- ◉ Types
- ◉ Manifestations
- ◉ Managements
- ◉ Complications

◎ Definition:

is a break or interruption in the continuity of a bone



❖ occur **mostly** from **trauma** but can result from a **pathologic process** in which bone has **degenerated**, such as in **osteoporosis** (metabolic bone disorder that causes a decrease in bone mass).



Types of Fractures

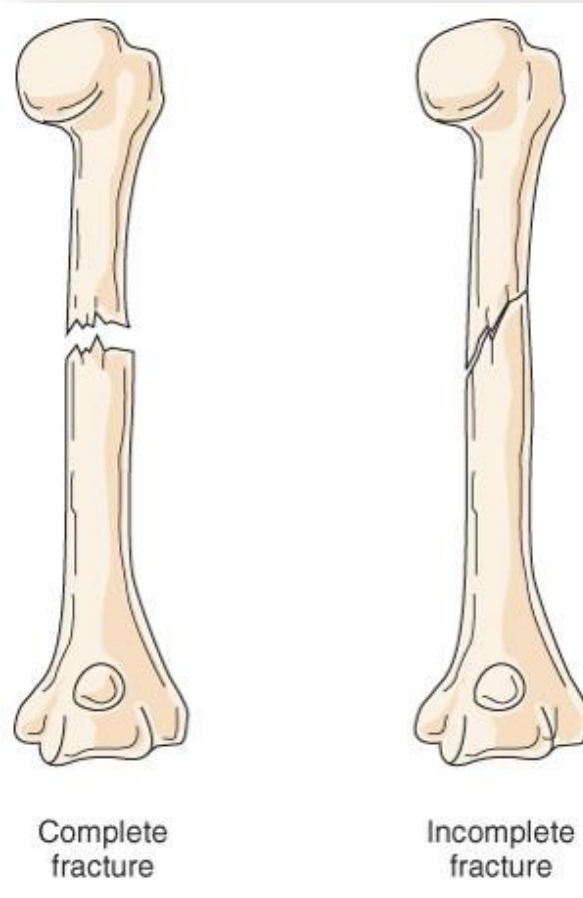
Complete & incomplete fracture

Closed & open fracture

Comminuted fracture

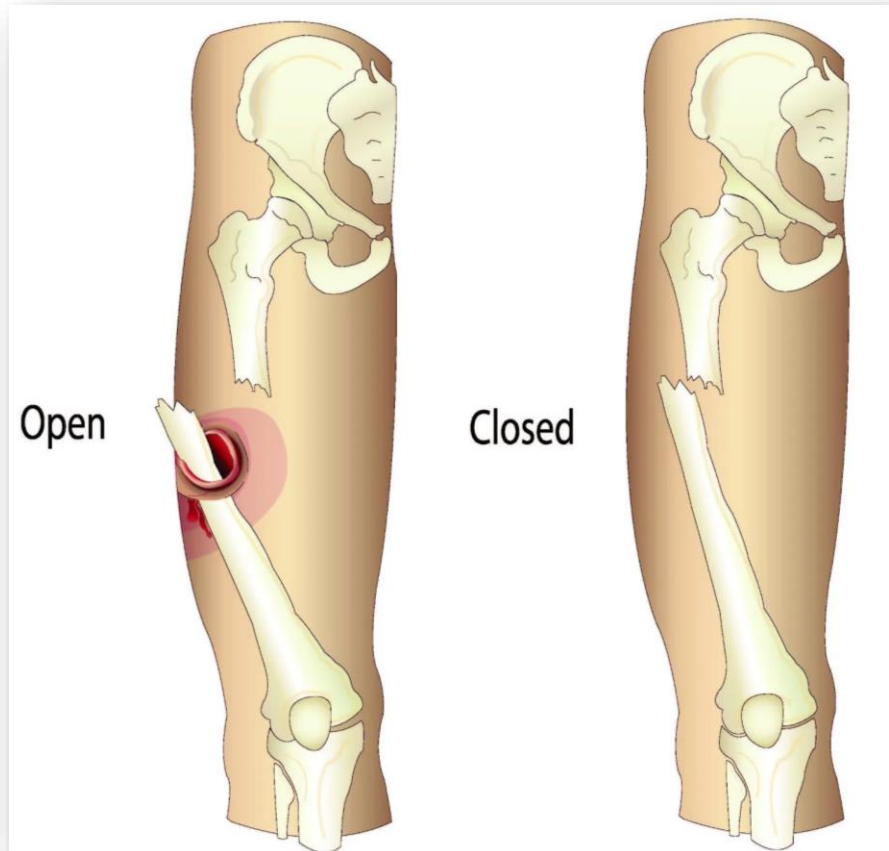
Greenstick fracture

- Complete fracture: bone breaks into two parts that are completely separated.
- Incomplete fracture: bone breaks into two parts that are not completely separated.



Closed fracture: in which there is no break in the skin.

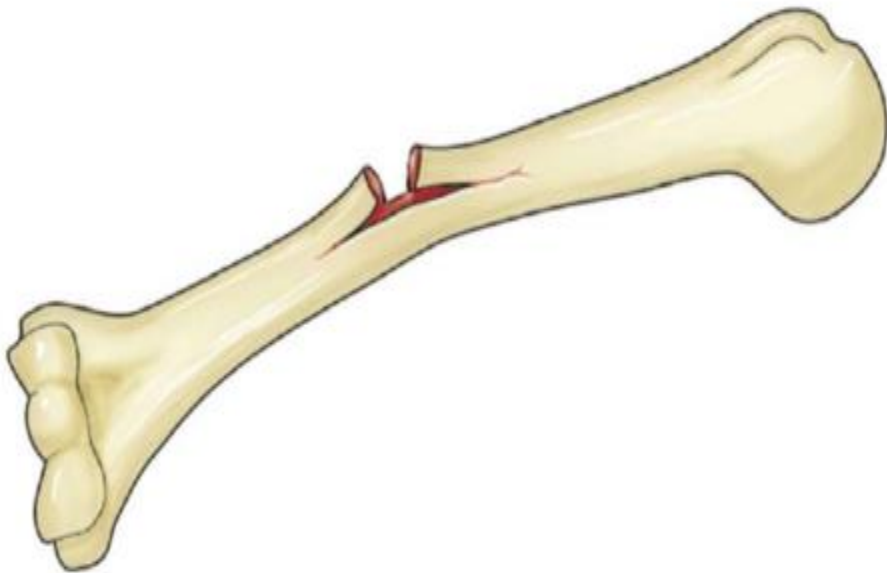
Open fracture: there is a break in the skin in which broken bone protrude.



Comminuted fracture: in which the bone is broken and shattered into more than two fragments.

Greenstick fracture: the bone is partially bent and partially broken

Greenstick



Comminuted Fracture



Manifestations

Pain (from mild to severe pain)

Its severity depends on type of fracture, the bone(s) involved, and the amount of displacement.

Tenderness

Deformity of the bone

Ecchymoses

Crepitation with any movement, and loss of function.

Symptoms of a Fractured Bone

SUDDEN PAIN



REDNESS



TROUBLE MOVING



DEFORMITY

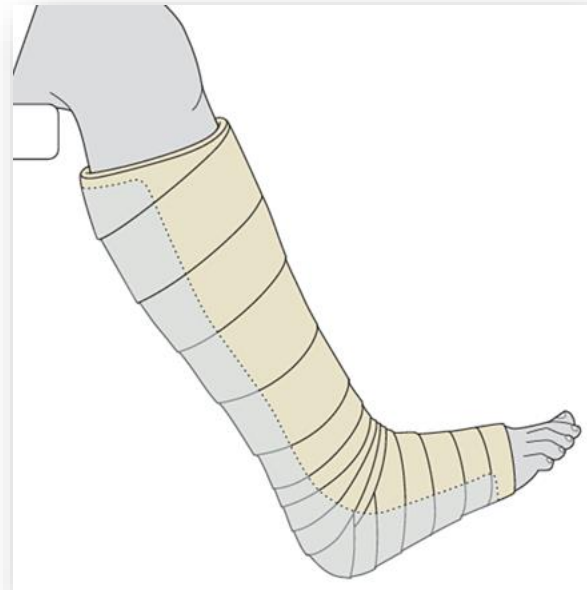


SWELLING



Management

Immediate **immobilization** (**Splint or cast**) of the part to prevent unnecessary damage. Immobilized in the **position** in which it is found at the time of injury.



The limb should be **supported** firmly so that it will **not be jarred** when the patient is being moved.



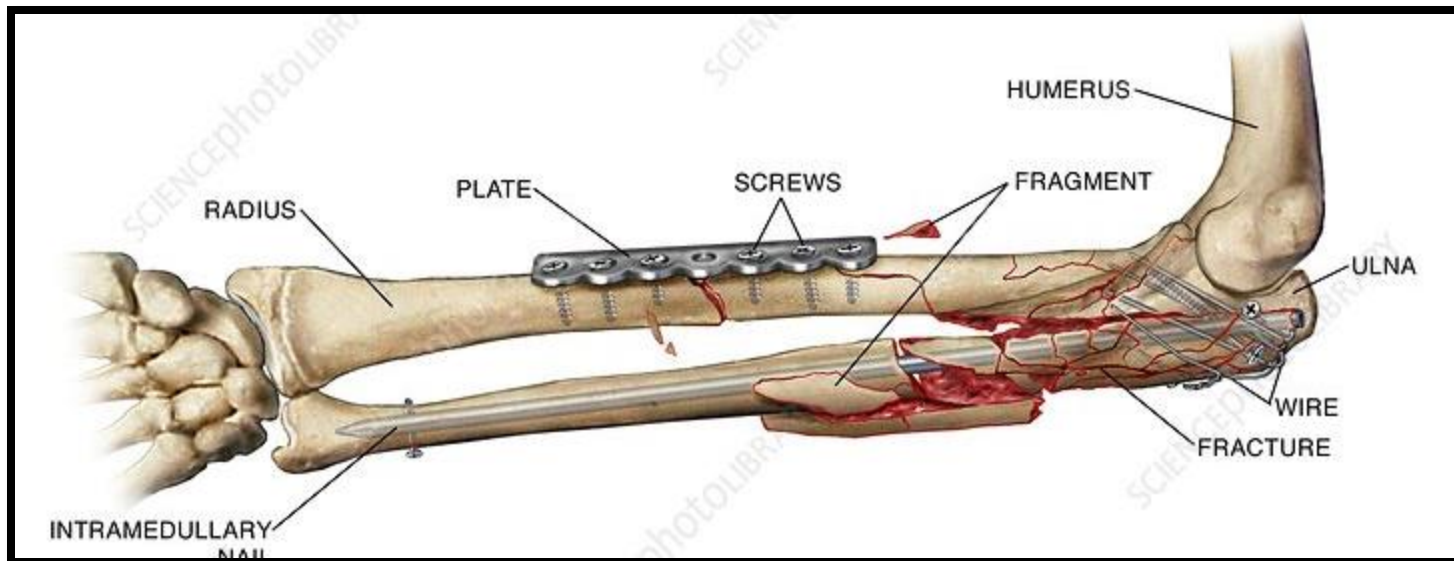
In open fracture , prophylactic antibiotics are usually administered

Ice in a plastic bag be applied to the fracture area to help minimize swelling, and decrease pain

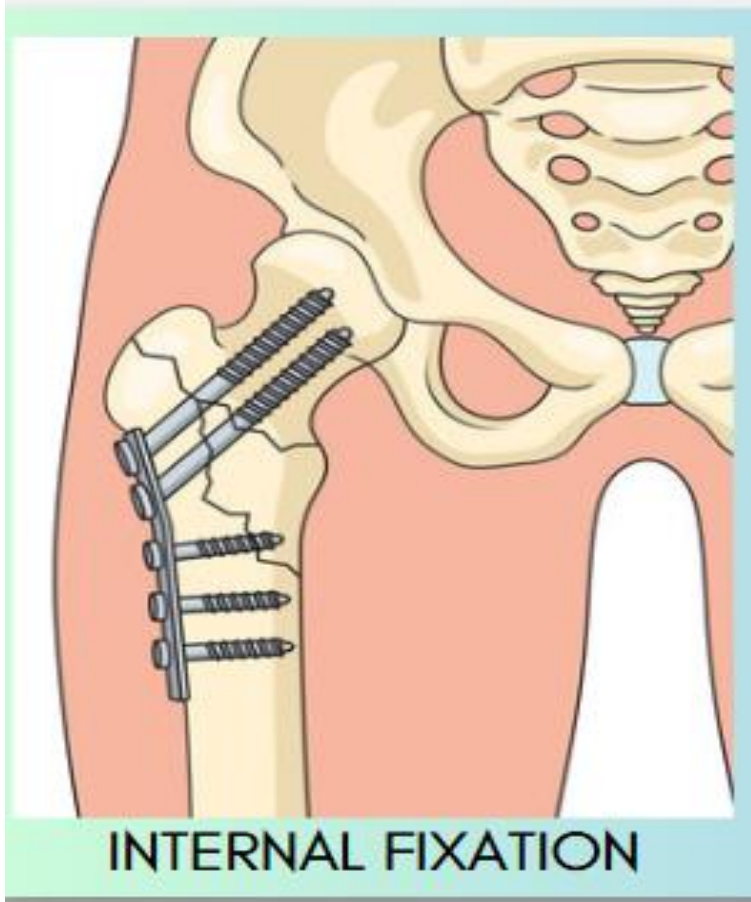


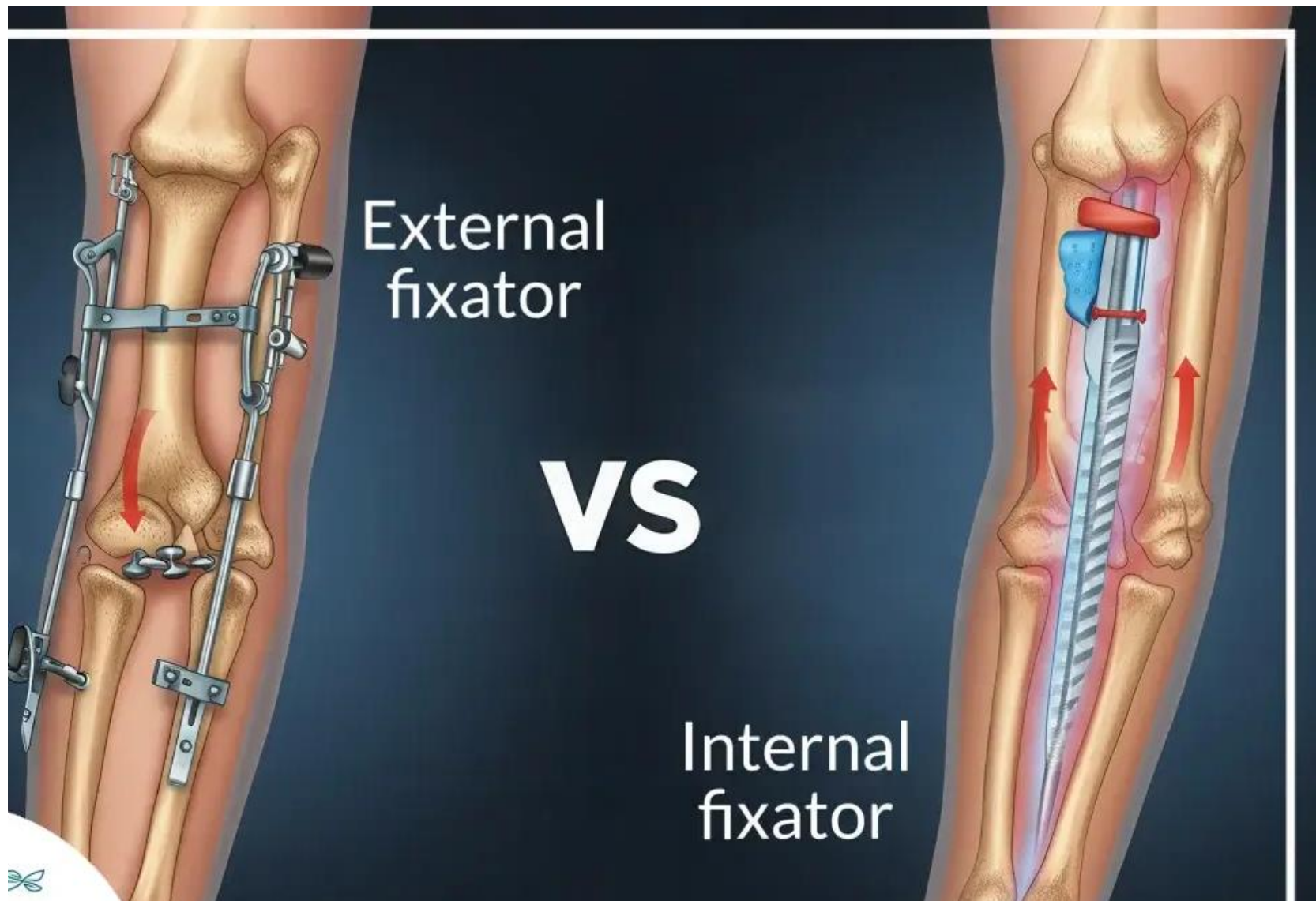
Surgical management

Used for severe fractures where an incision is made access the bone fragments, often combined with **Internal Fixation** which uses screws, plates to stabilize the bone.

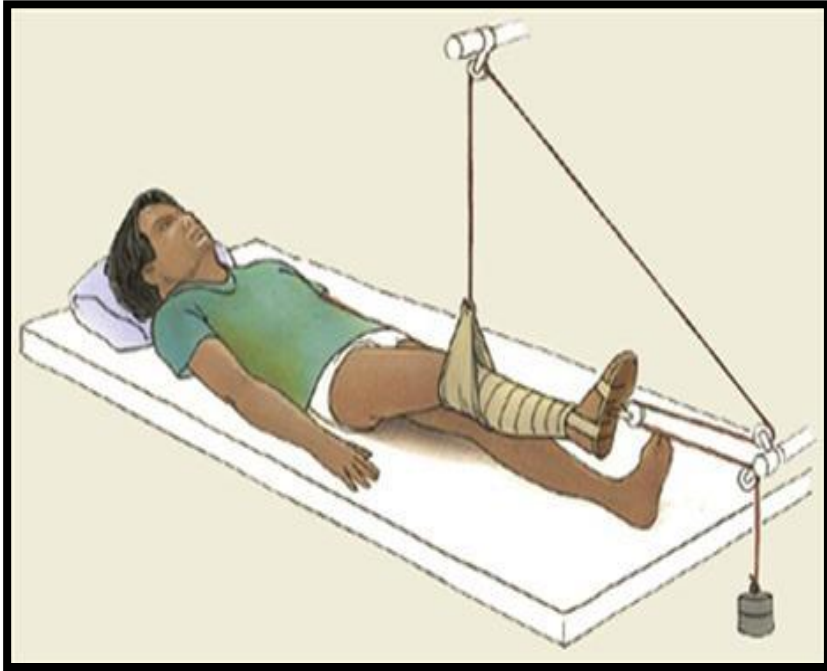


Types of fixation





Traction



Skin traction



Skeletal traction

CARE OF PATIENT IN TRACTION

**T
R
A
C
T
I
O
N**

Temperature

Extremity
Infection

Ropes Hang Freely

Alignment

Circulation Check (5 P's)

Type & Location of Fracture

Increase Fluid Intake

Overhead Trapeze

No Weights On Bed Or Floor

Complications of Fractures

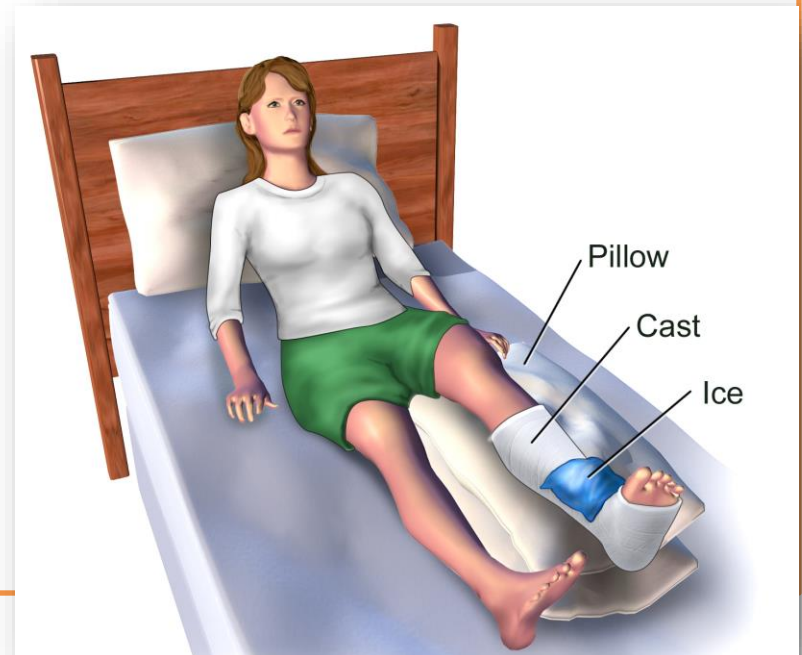
- **Hemorrhagic shock**
- **Damage to nerves from fragments of bone**
- **Fat emboli**
- **External pressure** from dressings or casts that are too tight
- **Infection**

Nursing managements

6 P Assessment

- Pain
- Pulse
- Paresthesia
- Pallor
- Paralysis
- Pressure

- Put patient in a **high Fowler's** position
- Establish a peripheral **IV** fluids.
- **Elevation** of the affected part to prevent swelling , toes should be higher than the trunk.



If a cast is in place, be **bivalved** (split through all layers of the material). **Dressings** will be **cut** or **replaced**



Wash hands

Use **aseptic technique** when caring for wound

Report **excessive drainage** or inflammation to physician

Culture drainage

Frequent monitoring of **temperature**





Summary



- **List 3 types of fracture**
- **List 4 clinical manifestations of fracture**
- **List 2 complications of fracture.**



Thank
you